

SpaceX Falcon 9 Data Science Capstone

Automated project report — generated

Executive Summary

This project analyzes SpaceX Falcon 9 launch data. Data was obtained from the IBM course dataset when the SpaceX API was unavailable. Exploratory data analysis, SQL queries, an interactive Folium map, a Plotly Dash skeleton, and multiple classification models were completed. Results include visualizations and model evaluation metrics; next steps are hyperparameter tuning and publishing to GitHub.

Introduction / Problem

Objective: Predict landing success (Class) for Falcon 9 launches and explore payload, launch site, and orbital patterns. Dataset: processed SpaceX launches CSV with 56 records used for demonstration.

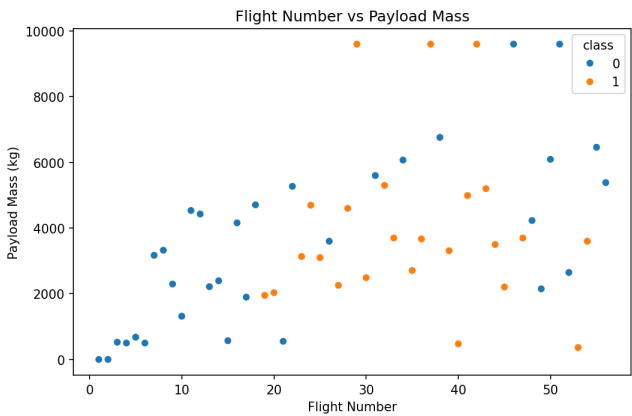
Data Collection & Wrangling

Source: attempted SpaceX REST API (v3/v4) but received server errors; used IBM course CSV copy saved to data/raw. Wrangling steps: parse dates, create Year, standardize launch site names, extract payload mass and booster version, and create `class` binary outcome.

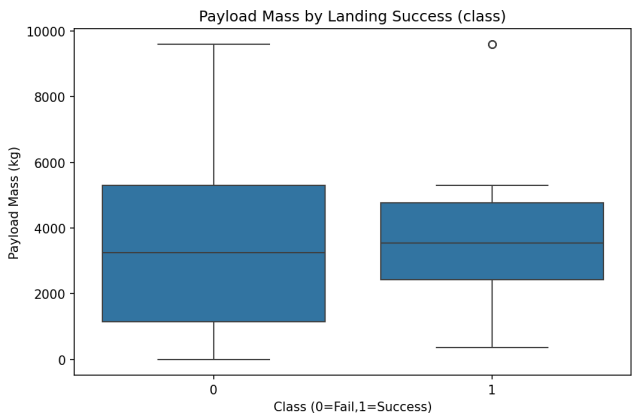
Web Scraping

Attempted to scrape Wikipedia for additional launch tables but received HTTP 403 Forbidden. A polite scraper with headers and retries was implemented in `src/web_scraping.py`; consider providing network allowance or manual download for full scraping.

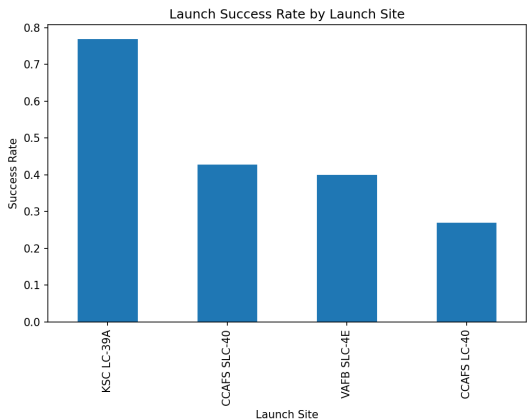
Exploratory Data Analysis



Flight Number vs Payload Mass



Payload Mass vs Landing Success



Launch Success Rate by Site

EDA with SQL (sample results)

SQLite queries could not be executed or DB missing

Folium Map

Interactive map generated at [figures/map/spacex_launch_map.html](#)

To include a static image in slides, open the HTML and capture a screenshot, then place it under [figures/map/](#).

Plotly Dash App

Dash app skeleton with dropdown, pie chart, payload slider, and scatter plot at `dashboard/app.py`

Run locally: `python3 dashboard/app.py`

Predictive Analysis — Model Results

LogisticRegression: accuracy=0.667, precision=0.500, recall=0.500, f1=0.500

SVM: accuracy=0.583, precision=0.333, recall=0.250, f1=0.286

DecisionTree: accuracy=0.750, precision=0.600, recall=0.750, f1=0.667

KNN: accuracy=0.583, precision=0.333, recall=0.250, f1=0.286

Conclusion and Next Steps

Findings: Launch success varies by site and booster version; payload distributions differ between successes and failures. Models show modest predictive performance on this small sample dataset. Next steps: hyperparameter tuning, additional features (orbit, core serial), more data, and publishing to GitHub.

GitHub Repository Status

No GitHub repository URL was provided or detected in this environment.

To include your GitHub URL in the final submission:

- Initialize git in project root: `git init`
- Add files: `git add .`
- Commit: `git commit -m "Initial capstone project"`
- Create repo on GitHub and push, then update README with the repository URL.